

Sample Problem Statements

1. Waste Sorting Assistant

SDG 12 - Responsible Consumption and Production

A single-agent solution that advises users on the correct disposal method for a given item based on its material and the user's local recycling rules. The agent uses a tool to look up disposal categories and applies memory to retain region-specific rules across interactions, improving accuracy over repeated use. Suitable as an introductory example of tool integration and memory management.

2. Energy Optimisation Assistant

SDG 7 - Affordable and Clean Energy

A multi-agent solution that helps households reduce electricity consumption. An intake agent gathers information about appliances and usage habits, an analysis agent estimates consumption and identifies high-impact areas using a calculation tool, and a recommendation agent produces prioritised, actionable savings tips. Demonstrates a coordinated multi-agent workflow and tool integration within a clearly bounded scope.

3. Water Usage Tracker

SDG 6 - Clean Water and Sanitation

A single-agent solution that records a user's daily water-related activities, estimates total consumption, and tracks usage over time. The agent relies on Agent Kernel's memory and session management to maintain a running record across multiple interactions and offers tailored suggestions for reducing consumption. A focused example of stateful, session-based agent behaviour.

4. Volunteer Matching Platform

SDG 17 - Partnerships for the Goals

A multi-agent solution that connects individuals with suitable volunteer opportunities. A profile agent captures the user's skills and availability, and a matching agent retrieves and ranks relevant opportunities from a provided dataset using a tool, returning the result with a brief justification. Memory/knowledge is used to retain user preferences for

future sessions. Demonstrates a coordinated multi-agent workflow, tool integration, and memory management.

5. Plain-Language Health Information Assistant

SDG 3 - Good Health and Well-being

A single-agent solution that explains common health and wellness topics in accessible language. The agent applies Agent Kernel's guardrails to remain strictly informational and to refer any clinical or diagnostic queries to a qualified professional. A minimal example of safe, guardrail-bounded agent design that requires no external integrations.